Task 2: 22 of 24 DF choices confirmed

This is selection validation, not the new heldout study. At the fixed plan-252 cut, 28 of 33 guided model/condition choices are frozen: 22 DF + 3 original QGF + 3 original FK. The two seed-37 stochastic averaged-Q conditions and all three original IPG choices remain pending. No new main heldout-100 episode has run.

Condition	lambda	Model seed 11	Model seed 23	Model seed 37
Regular / raw Q	0	35/40	40/40	37/40
Regular / raw Q	1	28/40	40/40	40/40
Stochastic / raw Q	0	40/40	40/40	39/40
Stochastic / raw Q	1	37/40	40/40	40/40
Regular / averaged Q	0	38/40	40/40	37/40
Regular / averaged Q	1	25/40	40/40	40/40
Stochastic / averaged Q	0	40/40	40/40	pending
Stochastic / averaged Q	1	35/40	40/40	pending
Original QGF	-	40/40	40/40	40/40
Original FK	-	40/40	40/40	40/40
Original IPG	-	NA: gate not qualified	NA: gate not qualified	NA: gate not qualified

Every displayed score is the selected candidate's successes among 40 fresh confirmation contexts. A candidate was selected from its literal overall top three after the declared earlier screening/VAL stages. Pending and unqualified cells are not zero-success results. These are three trained models with paired contexts, not a large independent-seed benchmark.

What these results support

Seed 23 reaches 40/40 for all eight DF conditions, QGF and FK, so success is at a ceiling there. Seed 11 and seed 37 show differences, but each row uses its own validated coefficients. This table does not show that deterministic or stochastic DF is better at the same kappa, eta and I, and it does not establish a causal benefit from averaging.

Closed native VAL/CONF executions	Executed native action rows	Closed worker time; parent clocks once	New main heldout
29,280	15,559,694	40,157.034103 s (11.155 h)	0

The execution count spans many settings, tasks and probes; reset contexts are paired or reused within declared roles. It is not 29,280 independent starts. Active seed-37 stochastic averaged-Q lambda0 VAL22 is excluded from these closed counts. [E1-E3]

Exact frozen settings, with inactive fields left blank

All DF rows use the unchanged full F8 endpoint and identity parameterization at temperature 1. Lambda is the latent-prior coefficient (0 or 1), not L1 regularization. M is inference flow slices; I is the DF inner iteration count; K is the fixed action-neighborhood bank size. FK N is particle count. A dash denotes an inactive field, not zero.

Seed	Method / Q	lambda	kappa	eta	I	sigma	K	M	N	FK noise	CONF40
11	Regular raw	0	0.1	0.1	256	-	-	8	-	-	35/40
11	Regular raw	1	40	0.03	256	-	-	8	-	-	28/40
11	Stochastic raw	0	5	0.1	128	-	-	8	-	-	40/40
11	Stochastic raw	1	5	0.1	128	-	-	8	-	-	37/40
11	Regular averaged	0	1	0.1	256	0.003	16	8	-	-	38/40
11	Regular averaged	1	64	0.03	256	0.5	4	8	-	-	25/40
11	Stochastic averaged	0	5	0.1	128	0.003	16	8	-	-	40/40
11	Stochastic averaged	1	5	0.03	128	0.003	16	8	-	-	35/40
11	QGF	-	16	-	-	-	-	64	-	-	40/40
11	FK	-	80	-	-	-	-	64	32	0.02	40/40
23	Regular raw	0	5	0.1	256	-	-	8	-	-	40/40
23	Regular raw	1	64	0.03	256	-	-	8	-	-	40/40
23	Stochastic raw	0	80	0.03	256	-	-	8	-	-	40/40
23	Stochastic raw	1	80	0.01	128	-	-	8	-	-	40/40
23	Regular averaged	0	16	0.1	256	0.3	4	8	-	-	40/40
23	Regular averaged	1	80	0.03	128	0.5	16	8	-	-	40/40
23	Stochastic averaged	0	64	0.1	128	0.5	4	8	-	-	40/40
23	Stochastic averaged	1	80	0.1	256	0.01	4	8	-	-	40/40
23	QGF	-	320	-	-	-	-	8	-	-	40/40
23	FK	-	80	-	-	-	-	32	32	0.02	40/40
37	Regular raw	0	0.1	0.1	256	-	-	8	-	-	37/40
37	Regular raw	1	5	0.1	128	-	-	8	-	-	40/40
37	Stochastic raw	0	5	0.1	128	-	-	8	-	-	39/40
37	Stochastic raw	1	5	0.1	128	-	-	8	-	-	40/40
37	Regular averaged	0	0.1	0.1	256	0.003	4	8	-	-	37/40
37	Regular averaged	1	5	0.03	256	0.003	4	8	-	-	40/40
37	QGF	-	16	-	-	-	-	16	-	-	40/40
37	FK	-	80	-	-	-	-	16	32	0.02	40/40

CSV companions: SELECTED22_DF_TUPLES.csv contains all 22 DF tuples. ALL_SELECTION_STATUS_AND_SETTINGS.csv retains all 33 Task-2 cells plus the three selected Task-5 QGF settings, full config JSON, selection paths and SHA256 commitments. No parameter was reranked by report generation. [E2-E4]

Harder problems: observed failures and honest limits

Task 5 native PointMaze selected QGF	kappa	M	Fresh CONF40
Model seed 11	96	16	40/40
Model seed 23	2	64	36/40
Model seed 37	80	32	40/40

Task 5 goes nominally from (20,4) to (0,0) in the maze, with native start noise and a 1,000-control-step cap. All 324 M/kappa cells were evaluated on VAL20, then nine literal top-three candidates received fresh40. The selected seed-23 policy has four failed confirmations; the saved diagnostic retains all 120 represented episodes. This is an observed navigation failure, not proof that the learned Q is wrong or that unseen actions caused it.

Paper Cube tasks remain a separate, untested comparison

Our PointMaze Task 2 / Task 5 labels are not the paper's Cube task numbers. The local CubeTriple Task 3 pilot has 46 state coordinates, physical action dimension 5 and H5 chunks (25 coordinates). It uses a limited 1M public slice and 128x128 networks, rather than the paper's 100M setting. No QGF-versus-eight-DF Cube native comparison has run.

The official registered Task-3 reward is pre-action: r = number of named cubes within .04 m of their goals - 3. The frozen million valid tuples have rewards -3:927,047; -2:71,460; -1:1,493; 0:0. Partial-goal signal exists, but joint success and goal-Q accuracy are unsupported by those observations. Goal-agnostic BC can learn useful primitives without a mandatory full-goal success cutoff.

Canonical checkpoint engineering passed all 206 predicates using six disposable BC calls (four effective + two repeated), including a published midpoint disk restore. It provides **zero baseline training credit**. At this cut, no fresh 50,000-update BC training or subsequent goal-IQL training is complete or released. [E5]

High-dimensional synthetic engineering: bounded partial, not a utility result

Saved cases	PASS / FAIL	Unrun	Actual API / data scope
548 of 596	538 / 10	48: whole D128 legitimate-narrow-good cell	1,116 public + 440 reference B20 calls; 96 separate fixture callbacks
All 20 D2 prerequisites + 48 fixture comparisons	PASS	No pilot adopted	548 NPZ / 13,048 arrays / 227,054,400 declared array bytes

The ten failures are original QGF at kappa80: the action-formula comparison exceeds unchanged 2e-6 absolute/relative tolerances; saved finite, bounds, replay and source checks pass. These are numerical reference-parity failures, not evidence that QGF performs worse on a task. The soft-deadline stop used 541.869712 parent seconds once; the generic unknown-tail flag stays true. No resume, pilot, IPG or old Task-5 failure waiver follows. The custom right-start Task-1 coverage probe likewise remains separate: its six best QGF cells reached 20/20, so it did not establish the sought QGF-failure case. [E6-E7]

Protocol, remaining barriers and reproducible evidence

Low-to-high grids and what was physically evaluated

Stage / method	Declared scope	Interpretation
DF TRAIN screening	20 kappas: 0,.02,.04,.08,.1,.2,.4,.8,1,2,3,5,8,12,16,20,32,40,64,80; 6 etas .001.. .1; l=1,2,4,8,16,32,64,128,256	All three models completed the fixed 9,240-setting hierarchy each. The three TRAIN16 copies represent one shared paired panel, not 48 independent observations.
Raw / averaged native DF	Raw shortlist <=7; averaged shortlist22. Literal top3 then fresh40. Averaged sigma=.003,.01,.03,.06,.1,.2,.3,.5; K=4 or16	TRAIN scores are not native success. Native evaluation covers the frozen shortlist, not the full kappa x eta x l Cartesian grid. Literal DF kappa0 still executes.
Original QGF native grids	27 ascending kappas 0..320 x M=8,16,32,64, per model	Task2 reuses 81 audited M8 B20 records once plus243new larger-M B20; Task5 executes all324new. Changing inference M does not retrain the F8 actor.
Matched curves	Declared same-kappa20 curves use independently validated fixed non-kappa anchors	Eta/l may differ between methods. Three existing raw seed11 overlaps also match eta/l; a full matched eta/l native grid has not been run.

Frozen score and selection roles

All nine PointMaze actor/Task2/Task5 configs use two target-Q heads aggregated by **minimum**. Raw Q is this frozen learned IQL target critic, not true return. Averaged Q is the finite fixed clipped antithetic action-neighborhood average Qbar(s,a) = (1/K) sum Q(s, clip(a + sigma epsilon_k,-1,1)); sigma is not the stochastic latent-update noise. BC, IQL and sampler coefficients are not trained online here.

Task2 roles remain distinct: selection reset IDs 8710001..20, confirmation IDs 8711001..40 and heldout IDs 8712001..100, with separate policy seeds. Own-best selection uses fresh40 only after the declared earlier stages. The all-33 barrier is still missing two DF choices and three unqualified IPG choices. l512 remains excluded by its retained failed engineering gate. The most recent independent IPG reference gate also retained 134 PASS / 10 FAIL; no learned Stage C or native IPG is admitted.

Next / outstanding	Scope at the fixed cut
Main DF	Seed37 stochastic averaged lambda0 VAL22 active; its CONF40 and lambda1 VAL22/CONF40 pending. No automatic next phase or concurrent DF branch.
Hard-problem readiness	Cube training/goal-Q/native H5 checks need separate actual completion, audits, cost and root release. Synthetic diagnostics do not qualify real learned models or universal RNG/state behavior.
Resources / handoff	Two local CPU numerical workers max; 4 GB/job; 24-hour prospective tranche. Closed40,157.034103 s; active7,200 s commitment; remaining39,042.965897 s. The verified private GitHub handoff freezes plan179 and is older than this report.

Evidence map: E1 PLAN252.json and owner cut; E2 all selected DF markers/metrics/CSV plus independent phase audits; E3 original QGF/FK choices; E4 exact tuple CSVs and all9 Task5 outcomes; E5 actual canonical audit4d8e9c25 and official reward addendumce7ab1e1; E6 highdim partial audit0a1b3bb4; E7 source/config/cohort receipts in EVIDENCE_MANIFEST.json. All reported scalars are in DATA.json. Full source hashes and copied evidence accompany the report. No model, array payload, sampler, RNG or native environment was queried to author it. This is an interim snapshot, not research closure.